

IKHLAQ ZAHIR

COMPUTER SCIENCE GRADUATE,
GAME DEVELOPMENT, HCI & AI-DRIVEN
INTERACTIVE SYSTEMS



CONTACT

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Portfolio:

SUMMARY

Computer Science graduate with experience in independently designing, developing, testing, and deploying games and Android applications. Development background spans Unity/C#, Kotlin/Android, algorithm design, procedural content generation. Seeking graduate research opportunities combining a software and game-development background with formal research in game technology, HCI, AI, interactive systems, or educational games.

EDUCATION

**Bachelors of Science in
Computer Science**
University of Management
and Technology, Lahore
2024

**Intermediate (pre-
engineering)**
Aga Khan Higher Secondary
School, Gilgit
2019

Matriculation
Aga Khan Higher Secondary
School, Gahkuch
2017

CORE COMPETENCIES

Game Development:

- Unity Engine, C#, 2D/3D Development
- Game Mechanics & Systems Design
- Game Architecture, Procedural Content Generation
- Constraint-Based Puzzle Design

Android Development:

- Kotlin, Android SDK, Jetpack Compose
- Android UI/UX, SQLite, JSON Processing
- In-App Billing, Google Play Deployment
- Mobile Application Architecture

AI & Machine Learning:

- Python, PyTorch, Machine Learning Fundamentals
- Generative AI, Large Language Models (LLMs)
- Transformer-Based Models, AI-Assisted Development
- On-Device/Lightweight AI

HCI & Interactive Systems:

- User Interface Design, UX-Oriented Development
- Interactive System Design, Player Experience
- Human-Computer Interaction, Interactive Learning Systems
- Game-Based Learning, User-Centered Design

CERTIFICATIONS

Android Game Development
in Unity 3D – PNY Trainings
May 2022

Unity 3D Game Design and
Development – Punjab Board
of Technical Education Lahore
October 2023

TOOLS AND
PLATFORMS

- Visual Studio
- VS Code
- Android Studio
- Unity
- Gradle
- SQLite

CORE COMPETENCIES

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ADDITIONAL INFO

Programming Languages:

C#, Kotlin, Python,
JavaScript/TypeScript

Game Engines:

Unity, Android Studio

Mobile Platform:

Android

AI/ML Tools:

PyTorch, Hugging Face, Google
Colab

Languages Spoken:

English, Urdu

PROFESSIONAL EXPERIENCE

Independent Game & Application Developer (2024 – current)

- Independently designed and developed multiple games and Android applications from concept through implementation, testing, iteration, and Google Play deployment.
- Game and application architecture; gameplay and interaction design.
- Programming, debugging, and algorithm development.
- UI/UX implementation and procedural generation.
- AI and gameplay systems; Android performance optimization.
- Database and data management.
- Testing, iteration, and Google Play release management.

PROFESSIONAL EXPERIENCE

- Game Development, Game Architecture, and Game Technology
- Game AI and Intelligent Interactive Systems
- Human-Computer Interaction (HCI) and User Experience
- Procedural Content Generation Level Design and environment Design
- Algorithmic Game Systems
- Educational and Serious Games

- Human-AI Interaction
- Interactive Learning Systems
- XR, Immersive Interaction, and Interactive Experiences
- Computational Game Design and Game Architecture

RESEARCH DIRECTIONS

Game Based Learning:

How interactive game mechanics can communicate complex scientific and technical concepts more effectively than conventional information presentation

Intelligent Game Systems:

How AI can adapt game experiences according to player behavior, knowledge, engagement, and ability

Procedural Content Generation:

How algorithms can generate meaningful game content while maintaining constraints, coherence, difficulty, and player experience

Human – AI Interaction in Games:

How AI agents can become interactive teachers, companions, NPCs, or collaborators within game environments

Interactive Simulations:

HCI and Immersive Experiences:

How novel interaction technologies, XR, and interactive environments can create more engaging and effective learning and gaming experiences

RESEARCH SKILLS

- Independent problem solving and algorithmic thinking.
- Technical prototyping and system design.
- Experimental thinking and iterative development.
- Translating concepts into working prototypes.
- Cross-disciplinary thinking spanning computing, games, AI, and HCI.